A Novel Radiometric Control Set Sample Selection Strategy for Relative Radiometric Normalization of Multitemporal Satellite Images

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***Abstract*— This article presents a new relative radiometric normalization (RRN) method for multitemporal satellite images based on the automatic selection and multistep optimization of the radiometric control set samples (RCSS). A novel image-fusion strategy based on the fast local Laplacian filter is employed to generate a difference index using the complementary information extracted from the change vector analysis and absolute gradient difference of the bitemporal satellite images. The difference index is then segmented into changed and unchanged pixels using a fast level-set method. A novel local outlier method is then applied to the unchanged pixels of the bitemporal images to identify the initial RCSS, which are then scored by a novel unchanged purity index, and the histogram of the scores is used to produce the final RCSS. The RRN between the bitemporal images is achieved by adjusting the subject image to the reference image using orthogonal linear regression on the final RCSS. The proposed method is applied to seven different data sets comprised of bitemporal images acquired by various satellites, including Landsat TM/ETM+, Sentinel 2B, Worldview 2/3, and Aster. The experimental results show that the method outperforms the state-of-the-art RRN methods. It reduces the average root-mean-square error (RMSE) of the best baseline method (IR-MAD) by up to 32% considering all data sets.**

***Index Terms*— Change vector analysis (CVA), multitemporal satellite images, radiometric control set samples (RCSS), relative radiometric normalization (RRN).**

I. Introduction

THE spectral information of the remotely sensed products usually has radiometric distortions such as variations in atmospheric conditions (e.g., atmospheric absorption and transmittance), sudden illumination changes, variations caused by the sun-target-sensor geometry, and the nonstability of the imaging sensors, which have a tendency to change over time [1], [2]. Thus, radiometric rectification is a necessary and crucial preprocessing step for multitemporal image analysis in remote sensing applications such as spatial–temporal change detection [3], unsupervised change detection [4], [5], time-series image analysis [6], UAS-based horticultural crop monitoring [7], image mosaicking [8], forest fires and burned area detection [9], deforestation [10], estimation of forest characteristics [11], drought monitoring [12], and coastal line monitoring [13]. In these applications, where a high level of automation and low computational cost with the least dependence on the *in situ* atmospheric data and sensor parameters at the time of imagery are required, relative radiometric normalization (RRN) methods are of interest [2], [14]–[17]. The RRN methods are based on adjusting the radiometric characteristics of a multiband subject image to a multiband reference image with the best visual quality and statistical robustness. These methods are generally comprised of two stages: 1) building a relationship [i.e., a radiometric mapping function (RMF)] between the subject and reference images and 2) producing the normalized subject image based on the generated RMF for each band in the previous step. It is worth noting that the subject and reference images are aligned spatially by an accurate coregistration method before applying the above steps.

The RRN techniques can be broadly classified into dense and sparse categories based on how they choose the pixels between the reference and subject images. The dense RRN (DRRN) approaches, such as simple regression (SR) [18], Haze correction (HC) [19], minimum–maximum (MM) and mean-standard (MS) deviation [14], and histogram matching (HM) [20], employ the whole of the reference-subject image pixel statistics to compute a mapping function for each band for adjusting the radiometric characteristics of the subject image. The DRRN methods form a linear RMF [14], [18], [19] or a nonlinear RMF [20] and produce satisfactory results given that the land-cover changes between the subject and reference images are negligibly small [10], [17]. Although they are computationally efficient, their performance significantly deteriorates if the land-cover change over the time is nonnegligible [16]. Furthermore, they are highly sensitive to noise. To alleviate the drawbacks of the DRRN

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methods, the sparse RRN (SRRN) approaches that employ invariant and stable radiometric control set samples (RCSS) selected from the subject and reference images are proposed. The selected RCSS are used to form an RMF for each band and the performance of the radiometric normalization is strongly dependent on the selection process [21], [22]. The manual selection of the RCSS is a time-consuming and tedious process; thus, the methods that automate the selection process are preferred [2], [17], [23]. For example, in the dark set-bright (DB) set method [14], the brightness and greenness features (which are generated by Kauth–Thomas transformation [24]) are thresholded to select the RCSS as the set of pixels that fall into "dark sets" and "bright sets." The pseudoinvariant features (PIFs) [23] that are based on man-made elements (e.g., concrete, asphalt, and rooftop) were used to select the RCSS based on the thresholding of the infrared-to-red ratio of the reference-subject image pair. Both methods employ linear regression to normalize each band of the subject image based on the selected RCSS. Although the PIF method generally produces better results than that of the DB method on the Landsat TM images, it still depends on the predetermined threshold values (similar to the DB method).

In [25], a fully automated SRRN approach is proposed, which automatically selects the RCSS using the principal component analysis (PCA) of the scattergram of the reference and subject images. The objective statistics based on the RCSS are used to create a linear regression model for the RRN. An automated quality control mechanism for the normalized image is employed to ensure the accuracy of the RRN. The main competence of this method is to guarantee the preservation of the radiometric resolution of the normalized subject image. In [26], the RCSS are selected as unchanged pixels between the reference and subject images using a spectral change-detection method. The RRN is achieved by applying linear regression to the automatically selected RCSS. The method highly depends on the performance of the change-detection method, which is based on the assumption that the spectral band differences between the reference and subject images follow Gaussian distribution. However, this assumption is too simple to model real data distribution over diverse land-cover changes. In [27], multivariate alteration detection (MAD) transformation [28] is used to select automatically the unchanged pixels as the RCSS. The linear regression is then applied to normalize the subject image. The method is simple and fast, yet produces satisfactory results when the temporal land-cover changes, especially seasonal vegetation changes, are minimal. Larger land-cover changes significantly reduce the fraction of the selected RCSS that deteriorate the accuracy of the method. To handle large land-cover changes, an iteratively reweighted modification of MAD (IR-MAD) transformation is used for the RRN [29]. The IR-MAD transformation performs significantly better than the MAD transformation in identifying the RCSS even under dominant temporal changes [29]. However, its performance is highly sensitive to the selected threshold value of probability for identifying the unchanged pixels. For instance, in the case of the reference-subject image pair with high spectral band differences, the threshold should be selected very close to 1 (e.g., 0.98), which may result in a sharp decrease in the fraction of RCSS that can affect the final RRN result. In [16], the unchanged pixels between the reference and subject images detected using a difference image and k-means clustering are used as RCSS. An artificial neural network (ANN) with one hidden layer is trained on a portion of the selected RCSS and used for the radiometric normalization of the subject image. The method can perform both linear and nonlinear normalizations. However, its computation time significantly increases with the increase in the number of selected RCSS. Furthermore, the ANN architecture employed, for the RRN plays a crucial role for the method's accuracy. The normalized difference water index (NDWI) along with the statistical rules is used for automatically normalizing multiple images with PIFs (MIPIFs) [17]. The experiments show that the method performs well in normalizing multiple images from a single sensor or multiple sensors. The method uses an empirical sampling rule (i.e., based on several numerical thresholds) for controlling the fraction of the selected RCSS that may not be applicable to different data sets with diverse land covers. Moreover, selecting only the unchanged dark and bright sets as the RCSS may not be suitable for all conditions (e.g., different dates and seasons).

In [30], a phenological RRN method was proposed for unsupervised change detection by taking the advantage of random forest (RF) regression and RCSS, extracted from the scattergrams between the near-infrared bands of the reference-subject image pair. Such a selection is based on the two assumptions including the lack of significant land-cover change in the data set and the selection of unchanged pixels with a correlation coefficient greater than 0.9. The experimental results showed the capability of RF regression compared with normalization based on adaptive boosting regression and stochastic gradient boosting regression. Nonetheless, it is possible that some of the suitable unchanged information might not be efficiently used through the one or both of the aforementioned assumptions. In [22], a semiautomatic SRRN method for high-resolution images was proposed considering the types of land covers. In this method, the RCSS are first selected based on the histogram of oriented gradient (HOG) features and the correlation coefficient and, then, they are divided into different categories using the object-based classification method. Although this method has relatively better RRN results, its performance significantly depends on the classification accuracy and the tuning parameters of the HOG algorithm (e.g., size of basic cells/blocks and block step). Moreover, its automation degree is considerably low due to the use of supervised classification.

In general, although preceding studies provided reasonable RRN results, there are still several limitations that should be addressed. For example, the current RRN methods do not provide an effective approach to refine the RCSS from the outliers, which may consequently result in an inaccurate model. In addition, most of the current approaches have limitations in properly dealing with data sets from different land covers. Furthermore, some of the presented RRN methods require manual selection of the detection thresholds to generate the initial and/or final RCSS. As other examples, in the SRRN methods developed based on the unsupervised change

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detection (see [16], [26]), a single operator (e.g., simple difference operator) is used to generate the difference index, which does not usually reflect the real changes and noise patterns in the study area. Moreover, in these approaches, a hard decision function (e.g., thresholding algorithm or k-means clustering) is used to discriminate unchanged samples as RCSS from the difference index, which leads to increase in uncertainty in the normalization process.

In this study, to cope with the limitations discussed above, a novel RCSS selection approach is proposed for the RNN of the multitemporal satellite images acquired over a wide range of land-cover changes. The proposed method automatically selects the unchanged samples and refines them step by step to generate the optimal RCSS. First, a novel fusion strategy based on the fast local Laplacian filtering (FLLF) is identified to generate an optimal difference index using auxiliary information from the change vector analysis (CVA) and absolute gradient difference (AGD) indices. Second, the unchanged samples are detected by applying the improved fast level-set method (IFLSM) [31] to the fused difference index. These samples are then refined based on a new local outlier-detection technique to identify the candidates. Third, a novel purity index is first introduced to score all of them and an automatic thresholding technique is then applied to the histogram of the candidate scores to extract the optimal RCSS. Finally, a normalized subject image is generated by exploiting orthogonal linear regression (OLR) to model the relationship between the reference and subject images through the optimal RCSS.

Our work mainly has four contributions, which includes the following.

1) Instead of using a single operator, a novel FLLF-based fusion strategy is proposed to generate the difference image that can reflect the real information of the changed regions and their geometrical details. Moreover, the potential of the IFLSM initialized by a fuzzy clustering is investigated to detect the unchanged pixels with lower uncertainty.
2) An outlier-detection technique based on the bivariate normal (BVN) joint distribution is proposed to refine the unchanged pixel pairs locally and help selection of the initial RCSS with appropriate spatial distribution in an image pair. This can be helpful in the accurate modeling of the relationship between the reference and subject images.
3) A novel purity index is introduced based on the probability of the no-change and the SSIM index, which can measure the reliability of the RCSS in various aspects, such as probability, important perceptual phenomena (e.g., luminance and contrast), and structure information (e.g., interdependences).
4) An iterative thresholding technique is developed based on the statistical parameters, which can extract the optimal RCSS in the fully automatic procedure, where the existing threshold-based SRRN methods lack in providing explicit and automatic strategies in selecting an appropriate threshold value.

The remainder of this article is organized as follows. Section II gives a brief review of the local Laplacian-based filtering methods and the fast level-set method (FLSM) including describing its use for binary segmentation. Section III describes our RRN approach, data sets, and the quantitative measures. Section IV provides the experimental results of the proposed RRN method, which are compared with the results of other state-of-the-art RRN methods in Section V. Finally, the concluding remarks are given in Section VI.

II. Background

A. Local Laplacian-Based Filtering Methods

The local Laplacian filtering (LLF) [32] is an edge-preserving and noise-reducing smoothing image algorithm that is based on the Laplacian pyramid decomposition, improved by Burt and Adelson [33]. The LLF involves three steps [34]–[36]: 1) the method applies a pointwise remapping function $r(.)$ that applies the Gaussian pyramid coefficient $g = G_\ell[D](m, n)$ [at level ℓ and position (m, n)] to the input image D, to generate multiscale remapped images $\{r(D)\}$s; 2) these images are combined to build each output coefficient $L_\ell[O](m, n)$ of the Laplacian pyramid; and 3) the output pyramid $L[O]$ is collapsed to rebuild the output image at the optimum scale. For all the coefficients of the output pyramid, this process should be repeated. Thus, remapping the image and computing a new pyramid for each value of g makes this procedure computationally expensive. To speed up this procedure, Aubry *et al.* [34] introduced an FLLF algorithm by linking the LLF to anisotropic diffusion and bilateral filtering. In this method, a small set of Laplacian pyramids are first precomputed at diverse values of g, where the intensity range is regularly discretized into several samples with these values. Then, the remapped pyramids for a particular g value are reconstructed by simple linear interpolation. The general form of remapping function is obtained as [34]

$$r(d) = d - (d - g) f(d - g) \tag{1}$$

where d is the intensity of image D and f is a continuous function. To simplify the calculation and generalize the LLF concept to the bilateral filtering algorithm, a Gaussian function is considered for f (e.g., for the detailed enhancement) [34]. Accordingly, the remapping function may be rewritten as [34], [36]

$$r(d) = d + (d - g) S_{\text{factor}} \exp\left((d - g)^2 / 2{\sigma_r}^2\right) \tag{2}$$

in which S_{factor} is the amplification factor that affects the smoothing of the details and σ_r is an intensity threshold parameter that describes the magnitude of the edges in the input image D.

B. FLSM for Unsupervised Change Detection

The FLSM is a conceptual algorithm and has been widely exploited in the image-processing field, especially in image segmentation [37]. This method is an improved version of the standard level-set method (LSM), which was introduced by Osher and Sethian [38]. The fundamental idea behind the LSM was to signify a closed contour (C) as the zero level set by a time-dependent PDE function $\varphi(t, m, n)$ as a level-set

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function [37], [38]. In image segmentation, as an example, the function φ divides the image space into two regions, including the detected pixels belonging to the objects and their boundary as the foreground and the remaining pixels as the background. In practice, the motions of the zero level set are formulated by converting the φ values with the velocity function F, which is expressed by the following elementary level-set equation [37], [39]:

$$\frac{\partial \varphi}{\partial t} + F|\nabla \varphi| = 0, \quad \varphi(0, m, n) = \varphi_0(m, n) \tag{3}$$

where $|\nabla \varphi|$ represents the normal direction, $\varphi_0(m, n)$ is an initial contour, and F denotes the comprehensive energy functions [39], [40]. Li *et al.* [37] introduced a new variation of the LS formulation as an FLSM, in which the costly reinitialization procedure is removed by forcing the LS function to be close to an Signed Distance Function (SDF). However, in the unsupervised change-detection approach, adopted in this article, such a model increases the risk of losing small changed regions and the edge of the regions with irregular boundaries in the LS evolution procedure [41]. To overcome these challenges, Moghimi *et al.* [31] extended the FLSM formulation for the unsupervised change-detection methods by adding two new terms to the total energy based on the standard deviation and mean of the changed pixels, as described in the following equation:

$$\frac{\partial \varphi}{\partial t} = \mu \zeta(\varphi) + \xi(\varphi, g_e, m_c, s_c) \tag{4}$$

where the function $\zeta(\varphi)$ is a penalty term to control φ deviation from the SDF, μ is a weighting coefficient of that, and the function $\xi(\varphi, g_e, m_c, s_c)$ is a term for controlling the movement of function φ toward the variational boundary of the changed regions. Accordingly, the function φ moves toward the boundary of the changed regions during an iterative process, which is expressed as follows:

$$\varphi^{k+1}(m, n) = \varphi^k(m, n) + \tau[\mu \zeta(\varphi^k) + \xi(\varphi^k, g_e, m_c, s_c)] \tag{5}$$

where φ^k and φ^{k+1} are the function φ in the k and $k + 1$ steps and τ is the time step of the LS evaluation, which can be determined by the change rate of φ. For a more detailed description of the FLSM method, adopted in this article, please refer to [31] and [37].

III. Materials and Methods

A. Proposed RRN Method

Let us consider two coregistered [42] optical satellite images, $R = \{r(m, n, i) \mid 1 \le m \le p, 1 \le n \le q, 1 \le i \le b\}$ and $S = \{s(m, n, i) \mid 1 \le m \le p, 1 \le n \le q, 1 \le i \le b\}$ as the reference and subject, with the size of $p \times q \times b$ ($p \times q$ pixels and b signify the number of spectral bands) acquired over the same geographical area at different times t_1 and t_2, respectively. The main objective of this study is to achieve the normalized subject image $\widehat{S}$, reducing the imaging condition effects (i.e., atmosphere and other effective parameters) based on reliable and well spatially distributed RCSS. The proposed RRN method is composed of three main phases (see Fig. 1) as follows.

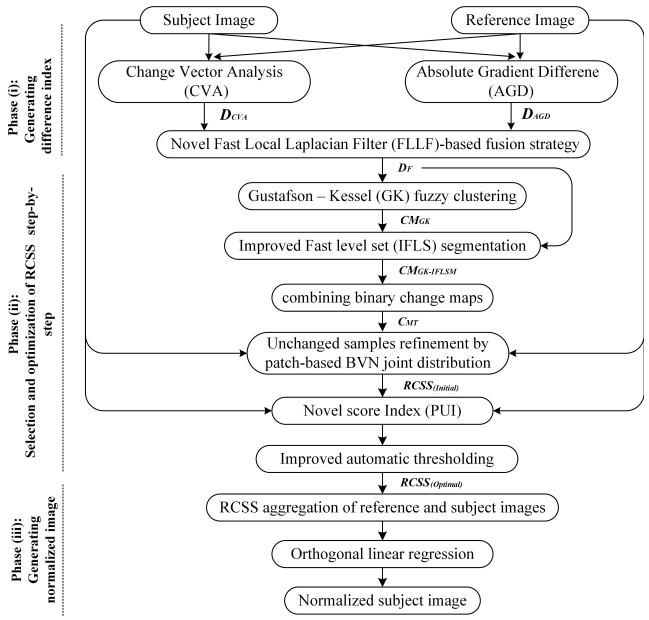

Fig. 1. Workflow of the proposed RRN method.

1) Producing an appropriate difference index from the subject and reference images using a novel FLLF-based fusion strategy.
2) Selection and optimization of the RCSS step by step are done in the following ways.
   a) Reducing the search space based on the elimination of the inappropriate samples (e.g., changed samples, noise-like patterns, and outliers) based on the IFLSM and a proposed local outlier detection.
   b) Selecting optimal RCSS by applying a novel automatic thresholding on the histogram of the candidate scores.
3) Generating the normalized subject image by rectifying the subject image to a reference image through a linear regression using the selected optimal RCSS. The details of each phase of the proposed algorithm are discussed in Sections III-A1–III-A3.

1) Phase (I) (Producing the Appropriate Difference Index): As shown in Fig. 1, the first phase of the proposed method is designed to generate an exclusive difference index $D_F = \{d_F(m, n) \mid 1 \le m \le p, 1 \le n \le q\}$ computed from reference image R and subject image S, which is employed for the further analysis of the RRN. First, two operators (i.e., CVA [43] and AGD) are exploited to create two difference indices from the input images R and S as follows:

$$D_{\text{CVA}} = \sqrt{\sum_{i=1}^{b} (R_i - S_i)}$$
$$D_{\text{AGD}} = \sum_{i=1}^{b} ||\nabla(G_\sigma * R_i)| - |\nabla(G_\sigma * S_i)|| \tag{6}$$

where $G_\sigma * R_i$ and $G_\sigma * S_i$ stand for the convolution of the reference and subject images with a smoothing Gaussian kernel G_s,

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respectively, and ∇ denotes the gradient operation. The CVA index provides a combination of the reflectance properties of the changed regions at different spectral bands [5]. Nevertheless, such a combination may cause a loss of weak boundaries of the changed regions. These details are highlighted in the difference index D_{AGD}, because the large gradient values can signify the details (e.g., edges and corners). However, this index does not produce information about the coverage of the changed regions. Based on the aforementioned, the use of a single difference index may not be sufficient to reflect the real information of the changed regions and their details. Hence, the information of the aforementioned indices can be combined to achieve an ideal difference index in which the noise, changed regions, and their geometric details are enhanced. Such an approach can facilitate their removal from the computational space in the next phase. To this end, we design a novel fusion strategy that can combine the indices based on FLLF [34] to highlight the spatial details and smooth the difference index simultaneously. To this end, the FLLF is first applied to the difference indices D_{CVA} and D_{AGD} to estimate the enhanced difference indices $O_{D_{\mathrm{CVA}}}$ and $O_{D_{\mathrm{AGD}}}$, respectively. Then, a low-level computing operator like the weighted mean operation is used to count the values of the fused index D_F as follows:

$$D_F(m,n) = \frac{D_{\mathrm{CVA}}(m,n)O_{D_{\mathrm{CVA}}}(m,n) + D_{\mathrm{AGD}}(m,n)O_{D_{\mathrm{AGD}}}(m,n)}{D_{\mathrm{CVA}}(m,n) + D_{\mathrm{AGD}}(m,n)} \quad (7)$$

where D_F is the fused difference index. To boost the performance of the fusion strategy, the values of D_{CVA}, D_{AGD}, $O_{D_{\mathrm{CVA}}}$, and $O_{D_{\mathrm{AGD}}}$ are normalized into [0, 1] before they are fused.

2) Phase (II) (Selection and Optimization of RCSS Step by Step): The second phase of the proposed method aims at giving a set of noise-resistant and stable RCSS using a step-by-step approach without any empiric sampling rule for controlling the fraction of the selected RCSS. This is performed by eliminating the unsuitable samples (e.g., changed regions, their geometrical details, and noise-like structures) and to reduce the search space for finding the time-invariant samples. In doing so, the IFLS segmentation [31] approach was first applied to the fused difference image D_F to detect the unchanged samples. As mentioned in Section II, the IFLS divides the difference index D_F into two zones: the change region inclusive of its boundary (i.e., $\varphi(m,n) \leq 0$) and the unchanged region (i.e., $\varphi(m,n) > 0$). It is also worth noting that the function φ requires an arbitrary initial contour for starting the contour evaluation. Here, the IFLSM directly uses the produced change map $\mathrm{CM}_{\mathrm{GK}}$ from applying Gustafson–Kessel (GK) fuzzy clustering [44] to the difference index D_F as an initial contour for automatic initialization and parameter configuration, which is obtained as follows:

$$\phi_0(m,n) = \begin{cases} 2\varepsilon, & \text{if } \mathrm{CM}_{\mathrm{GK}}(m,n) = 1 \\ -2\varepsilon, & \text{if } \mathrm{CM}_{\mathrm{GK}}(m,n) = 0 \end{cases} \quad (8)$$

where ε is a constant adaptable in the smoothed form of a typical Dirac function $\delta_\varepsilon(.)$, which is proposed in [37] and the changed and unchanged regions are denoted by

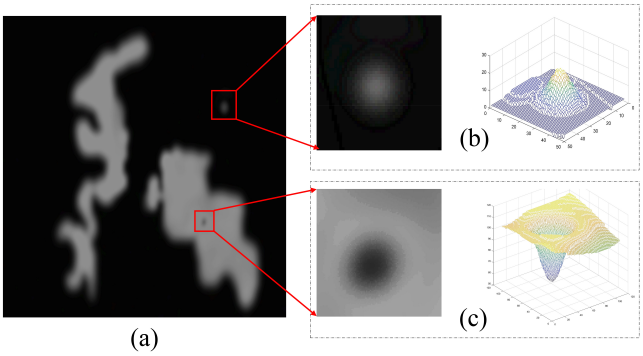

Fig. 2. Illustration of noise-like structures in a difference image. (a) Simulated difference image. (b) Noise-like type one (appears like a changed). (c) Noise-like type two (appears like an unchanged).

"0" and "1," respectively. After initializing the FLSM by $\mathrm{CM}_{\mathrm{GK}}$, the function φ moves toward the boundary of the changed regions during contour evolution according to (5) and the robust binary change map $\mathrm{CM}_{\mathrm{GK-IFLSM}}$ is achieved by terminating after a finite number of iterations based on the stopping criteria $|\varphi^{k+1} - \varphi^k| \leq 0.5$.

Here, we are looking for a change map, in which not only the changed regions are identified but also the types of noise and noise-like structures are detected. In general, we faced with two noise-like patterns in the difference index, and the patterns look like the changed regions named noise-like type one [see Fig. 2(b)] and another one that has a similar behavior to the unchanged regions named noise-like type two [see Fig. 2(c)]. In practice, these structures affect the final RRN results if they are not detected along with the changed regions.

As shown in Fig. 2, the noise-like type one looks like a pike out of the surface, which can be easily identified with the initial couture ($\mathrm{CM}_{\mathrm{GK}}$), where the noise-like type two remained. Moreover, the noise-like type two resembles a hole in the changed regions, which can be removed by expanding the contour of the IFLSM forward in each iteration, while the contour crosses from the noise-like type one. Accordingly, to detect the noise-like patterns, the combination of two change maps $\mathrm{CM}_{\mathrm{GK-IFLSM}}$ and $\mathrm{CM}_{\mathrm{GK}}$ was here used as follows:

$$CM_T = \mathrm{CM}_{\mathrm{GK}} \cup \mathrm{CM}_{\mathrm{GK\ -\ IFLSM}} \quad (9)$$

where $\cup$ performs the binary union operation and CM_T is the final change map. In this step, all the unchanged samples in the final change map CM_T are selected for the further analysis of the normalization procedure. In essence, this concept can boost the performance of the proposed method in removing the gross errors in the RCSS selection strategy.

It is expected that the detected unchanged samples in the reference and subject images are linearly related to each other. However, this linear relationship is not definite due to the existence of several unchanged pixel pairs with unreasonable values (outliers) [45]. To solve the problem, inspired by Teng *et al.* [45], we present a local outlier-detection technique that uses the BVN joint distribution method called patch-based BVN joint distribution to refine the unchanged pixel pairs, where a global outlier-detection technique can be an inefficient

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task when dealing with the unchanged pixel pairs with a variety of structures (depending on the type of unchanged land cover involved). In this technique, the reference and subject images are first masked by the change map CM_T to generate unchanged pixel pairs. Then, the masked image pair is divided into small nonoverlapping patches. Finally, the unchanged pixel pairs in each patch are individually refined by the BVN joint distribution technique. Let us consider R_{uc} and S_{uc} as the random variables of the unchanged pixels in each patch of image pair, respectively, both of which are assumed to have a BVN distribution if their joint probability density function (pdf) is

$$f(U) = \frac{1}{2\pi\,|\Sigma|^{0.5}} - 0.5[(U - m_{\mathrm{uc}})'\Sigma^{-1}(U - m_{\mathrm{uc}})] \quad (10)$$

where $U^T = (R_{\mathrm{uc}}, S_{\mathrm{uc}})$, m_{uc}, and Σ are the unchanged pixel pairs, mean vector, and covariance matrix of U for each patch, respectively. The patch-based BVN joint distribution considers the Mahalanobis distance between the DN values of the unchanged pixels in the subject–reference image pair. Such a distance is due to the radiometric deformation (e.g., atmospheric condition) between the acquisition dates [45]. In addition, this distance is known to have a Chi-square distribution with 2° of freedom [46]. Therefore, unchanged pixel pairs can be evaluated using the Chi-square distribution with significance level α as

$$P[(U - m_{\mathrm{uc}})'\Sigma^{-1}(U - m_{\mathrm{uc}}) \le \chi^2_{1-\alpha}] = c \quad (11)$$

where c is a probability contour. The outliers can be mainly detected using (11). However, the outliers also contribute to the generation of the Mahalanobis distance parameters (m_{uc} and Σ) [47], [48]. To overcome this limitation, we use the "minimum covariance determinant" method [46] to calculate parameters due to simplicity and the fast computing process. Finally, the refinement of the unchanged sample pairs for each patch can be formulated from a hypothesis test based on the probability contour c as

$$\begin{cases} H_0 : U = \text{unchanged} \\ H_1 : U = \text{outlier} \end{cases} \quad (12)$$

where H_0 is the null hypothesis that all unchanged pixel pairs fall within the probability contour c and are selected as initial RCSS. In addition, H_1 is the alternative hypothesis that refers to reject the null hypothesis. Such a process is implemented in each spectral band of the reference–subject image pair that can be seen as a decision of a member of a pool of experts. Thus, the unchanged pixels are allocated to the outlier class while obtaining the highest number of votes in H_1 for all spectral bands and the rest of the unchanged pixels are selected as the initial RCSS. As an example, the patch-based BVN joint distribution procedure conducted on the four patches of the reference and subject images is depicted in Fig. 3, where the unchanged pixel pairs in each patch have different scatterings, because the patches have a different mean and standard deviation that leads to the deformation of the probability contour in each patch.

In the next step, we proposed a novel purity unchanged index (PUI) to determine the degree of purity of the initial

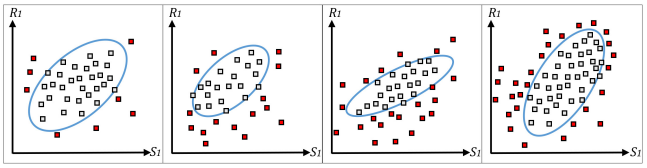

Fig. 3. Detecting outliers (red squares) from unchanged pixel pairs (white squares) on four patches with a 95% probability contour at $\alpha = 0.05$ (blue contour).

RCSS by scoring them in terms of their invariance properties, which can be expressed by a multiplicative combination of the two terms as

$$P_{\mathrm{UI}} = \sqrt{P_r(\text{unchanged}) \cdot \mathrm{SSIM}_{\mathrm{Map}}} \quad (13)$$

where $P_r(\text{unchanged})$ denotes the probability of no change, which is indirectly estimated by measuring the distance based on the quadratic discriminate function

$$P_r(\text{unchanged}) = -0.5 \ln|\Sigma_N| - 0.5(Z - m_N)'\Sigma_N^{-1}(Z - m_N) \quad (14)$$

where $Z = \{z_1, z_2, \ldots, z_b\}$ are the selected initial RCSS in the each band of the spectral change index ($\mathrm{SC} = |R - S|$), and m_N and Σ_N are their mean and covariance, respectively. In addition, $\mathrm{SSIM}_{\mathrm{Map}}$ is the mean of the local structural similarity index (SSIM) [49] between the reference R and subject S images, in which the initial RCSS are considered as

$$\mathrm{SSIM}_{\mathrm{Map}} = \sqrt{\sum_{i=1}^{b} \frac{(2\mu_{S_i}\mu_{R_i} + D_1)(2\sigma_{S_iR_i} + D_2)}{(\mu_{S_i}^2 + \mu_{R_i}^2 + D_1)(\sigma_{S_i}^2 + \sigma_{R_i}^2 + D_2)}} \quad (15)$$

where μ_R and μ_S, and σ_R and σ_S are the local means and standard deviations of the images R and S and σ_{SR} is the local cross-covariance. Furthermore, $D_1 = (0.01d_r)^2$ and $D_2 = (0.03d_r)^2$ are the nonzero regularization constants to avoid uncertainty in the image regions, where d_r is the dynamic range of pixel value (i.e., obtained by $(2^{\mathrm{bits}} - 1)$, where bits denotes to the radiometric resolution). Due to the occurrence of negative values in P_r and $\mathrm{SSIM}_{\mathrm{Map}}$, their values were normalized to the range of [0, 1]. Considering the probability criterion with a visual impact on the three characteristics of the subject images compared with the reference images (e.g., luminance, contrast, and structure) can provide a realistic estimate of the quality of the RCSS.

After estimating the PUI, we have the vector $P_{\mathrm{UI}} = \{p_1, p_2, \ldots, p_{N_p}\}$, which is a set of N_p (the number of initial RCSS) scores of the initial RCSS in the range [0, 1]. When the p value is closer to 1, a stronger value of the initial RCSS is indicated, which can be a good candidate for selecting as optimal RCSS. Here, an automatic iterative thresholding method is proposed based on the statistical parameters of the candidate scores to select the optimal RCSS as follows.

1) Select the difference between the median and median-based coefficient of variation of the PUI vector as an initial value for the threshold t.
2) Divide the histogram of the initial RCSS scores into two parts, PUI_1 (unreliable initial RCSS) and PUI_2 (reliable initial RCSS), using the threshold t.

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3) Calculate the median and the median-based coefficient of the variation values of the PUI_2 as

$$CV_{\mathrm{Med}}(\mathrm{PUI}_2) = \frac{\mathrm{Mad}(\mathrm{PUI}_2)}{\mathrm{Med}(\mathrm{PUI}_2)} \tag{16}$$

where Med(.), $\mathrm{CV}_{\mathrm{Med}}$(.), and Mad(.) are the median, the median-based coefficient of variation, and the median absolute deviation operators, respectively.

4) Compute the new threshold as

$$t = \mathrm{Med}(\mathrm{PUI}_2) - 2\mathrm{CV}_{\mathrm{Med}}(\mathrm{PUI}_2). \tag{17}$$

5) Repeat Steps 2–4 until the difference between the two consecutive threshold values becomes insignificant ($|t^{k+1} - t^k| \leq 5 \times 10^{-4}$, where k and $k+1$ are the kth and $k+1$ iterations, respectively).

The improved thresholding method is free of any distribution due to the utilization of the median absolute deviation operator of the scores in the $\mathrm{CV}_{\mathrm{Med}}$.

3) Phase (III) (Generating the Normalized Image): The third phase of the proposed method provides a normalized subject image by applying OLR [50] to the subject–reference image pair through the selected optimal RCCS. The OLR calculates error parallel to the reference image axis, $r_j = R_j + \epsilon_j$, and the subject image axis, $s_j = S_j + \upsilon_j$, where ϵ and υ are the measurement errors. Thus, such a regression method minimizes the sum of squares perpendicular distances from the collected optimal RCSS, $\mathrm{RCSS}_j(r_j, s_j) = (R_j + \epsilon_j, S_j + \upsilon_j)$, $j = 1, 2, 3, \ldots, N_o$, (N_o is the number of optimal RCSS) to the fit line L in each spectral band of the images as

$$E_{OR} = \sum_{j=1}^{N_p} d(\mathrm{RCSS}_j, L)^2 = \sum_{j=1}^{N_p} \frac{(r_j - \beta_1 s_j - \beta_0)^2}{\beta_1^2 + 1} \tag{18}$$

where d is the deviation measured perpendicularly from the each of the RCSS to the fit line, called the residual. By minimizing the E_{OR} based on the least-squares method, the RRN coefficients (intercept coefficient β_0 and the slope coefficient β_1) are calculated as [51]

$$\beta_0 = \frac{\sum_{j=1}^{N_p} r_j}{N_p} - \frac{\beta_1 \sum_{j=1}^{N_p} s_j}{N_p}, \quad \beta_1 = \frac{A + \sqrt{A^2 + 4B^2}}{2B} \tag{19}$$

where

$$A = \left(\sum_{j=1}^{N_p} s_j\right)^2 - \left(\sum_{j=1}^{N_p} r_j\right)^2 - N_p\left(\sum_{j=1}^{N_p} {s_j}^2 - \sum_{j=1}^{N_p} {r_j}^2\right) \tag{20}$$

and

$$B = N_p \sum_{j=1}^{N_p} s_j r_j - \sum_{j=1}^{N_p} s_j \sum_{j=1}^{N_p} r_j. \tag{21}$$

Finally, using the RRN coefficients in each spectral band, the normalized subject image $\widehat{S}$ is calculated by the linear formula

$$\widehat{S_i} = f(S_i) = \beta_{0i} + \beta_{1i} S_i, \quad i = 1, 2, \ldots, b. \tag{22}$$

In summary, the main phases of the proposed SRNN method are presented as follows.

1) Generating the suitable difference index by the FLLF-based fusion strategy [see (7)].
2) The step-by-step RCSS selection and optimization are as follows.
   a) Detecting the unchanged samples by analyzing the fused difference image using IFLSM initialized by the GK clustering (GKC) result.
   b) Generating the initial RCSS by refining the detected unchanged samples from outliers using the patch-based BVN joint distribution method through (12).
   c) Computation of the proposed score index (PUI) for the initial RCSS through (13).
   d) Selection of the optimal RCSS by applying the proposed thresholding technique on the histogram of the candidate scores.
3) Generating the normalized subject image.
   a) Computation of the RRN coefficients based on the optimal RCSS by (19).
   b) Normalizing the subject image using (22).

B. Data Set Description

To demonstrate the efficiency of the proposed method for the RRN, seven coregistered bitemporal satellite images acquired from different geographical areas were assessed in this study. These data sets mainly contained different land covers (e.g., water body, urban region, vegetation, and so on) and are available at GitHub [52]. The false-color composite of the data sets and the corresponding normalized subject images generated by the proposed method are presented in Fig. 4.

Data set 1 is composed of a bitemporal Landsat 7 (ETM+) scene (300 × 400 pixels) over Lago Mulargia in the Province of Cagliari (Italy) acquired on September 5, 2002 (Subject image) and May 19, 2003 (reference image). Data set 2 includes two images (700 × 700 pixels) acquired by the Landsat 7 (ETM+) and Landsat 5 (TM) satellites on September 5, 1990 (subject image) and May 28, 2003 (reference image), respectively, over Palm Jumeirah in Dubai city (United Arab Emirates). Data set 3 is composed of two images (2000 × 2000 pixels) acquired by the Landsat 7 (ETM+) and Landsat 5 (TM) satellites on June 20, 1978 (subject image) and April 21, 2003 (reference image), respectively, over the Hamoun Lake in the Southwestern Farah Province/Northwestern of Nimruz Province (Afghanistan). Data set 4 includes a bitemporal IRS-P6 (LISS-3) scene (1100 × 1100 pixels) over Varzaqan City in the East Azerbaijan Province (Iran) acquired on July 8, 1998 (subject image) and May 8, 2007 (reference image). Data set 5 is composed of two images (2000 × 2000 pixels) acquired by the WorldView-2 and WorldView-3 satellites on June 9, 2011 (subject image) and July 7, 2015 (reference image), respectively, over the town of Bzin in the northwest of Shiraz Province (Iran). Data set 6 is composed of bitemporal Sentinel 2B scenes (2000 × 2000 pixels) over Haditha Dam in the Haditha city (Iraq) acquired on August 21, 2015 (subject image) and

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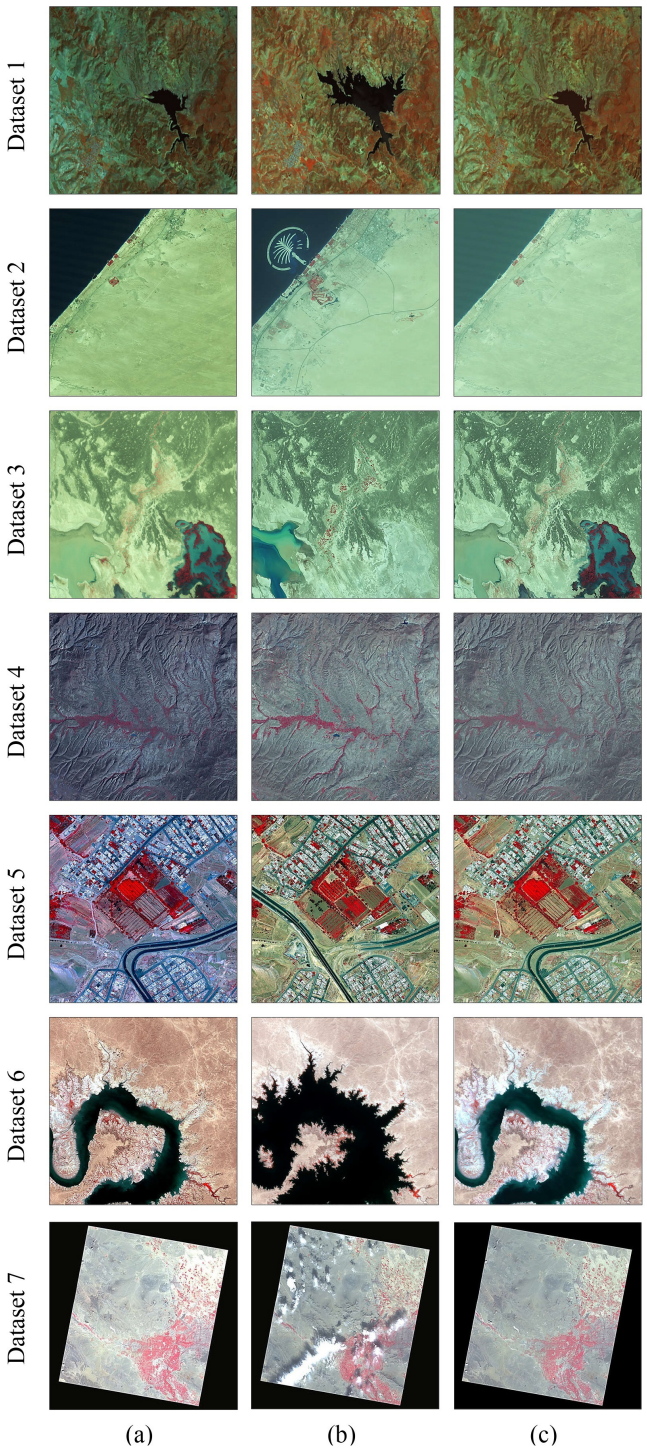

Fig. 4. False-color composites (NIR/red/green) of seven data sets and the corresponding normalized subject images. (a) Subject images. (b) Reference images. (c) Normalized subject images by the proposed method.

October 4, 2016 (reference image). Data set 7 comprises of a pair of the full-scene ASTER (4927 × 5575 pixels) over the West of the Isfahan city (Iran) acquired on March 27, 1998 (subject image) and September 29, 2007 (reference image).

The thermal bands (bands 10–14 of the ASTER images and band 6 of the Landsat 7/5 images) and the cirrus band (band 10 of Sentinel 2B images) were discarded for the RRN. The 30-m/pixel bands [six short-wave infrared (SWIR) bands] of the ASTER images were sharpened to the 30 m/pixel of the three visual near-infrared (VNIR) bands by the PCA fusion technique with the PCA-based pan-sharpening technique [53]. The spatial resolution of the 20- and 60-m/pixel bands of the sentinel 2 images was enhanced to 10 m/pixel by the Sen2Res model [54], which is available in the Sentinel Application Platform (SNAP) open-source software [55]. For data set 5, first, the spectral bands of the WorldView2 and WorldView3 images were sharpened to 31- and 40-cm/pixel panchromatic bands, respectively, with the PCA-based pan-sharpening technique [53]. Then, both of them were resampled to 40 cm/pixel by nearest-neighbor resampling.

C. Accuracy Estimation of RRN

Accuracy evaluation of radiometric normalization was performed using two approaches. First, the RRN methods were assessed to evaluate the accuracy of radiometric normalization in terms of the average root-mean-square error (RMSE) and the average relative mean absolute difference (RMD) on the test data as follows:

$$\mathrm{RMSE}_{\mathrm{Ave.}} = \sqrt{\frac{1}{n_t * n_b}\sum_{i=1}^{n_t}\sum_{b=1}^{n_b}\left(R_{i,b} - \hat{S}_{i,b}\right)^2} \tag{23}$$

$$\mathrm{RMD}_{\mathrm{Ave.}} = \frac{\sum_{i=1}^{n_t}\sum_{b=1}^{n_b}\left|R_{i,b} - \hat{S}_{i,b}\right|}{\sum_{i=1}^{n_t}\sum_{b=1}^{n_b} R_{i,b}} \times 100 \tag{24}$$

where n_t is the total number of test samples. The acceptable normalized image quality derived from the RRN is characterized by a low RMD and RMSE. In the second approach, in addition to the mean value (MV) and the standard deviation value (SV), three quantitative measures were used to evaluate the generated normalized image using the proposed method concerning the reference image, including the SSIM, relative RMSE (RRMSE), and peak signal-to-noise ratio (PSNR). These were calculated on the test data for each band as follows:

$$\mathrm{SSIM}(\hat{S}, R) = \frac{(2\mu_{\hat{S}}\mu_R + D_1)(2\sigma_{\hat{S}R} + D_2)}{(\mu_{\hat{S}}{}^2 + \mu_R{}^2 + D_1)(\sigma_{\hat{S}}{}^2 + \sigma_R{}^2 + D_2)} \tag{25}$$

$$\mathrm{RRMSE} = \frac{\sqrt{\frac{1}{n_t}\sum_{i=1}^{n_t}\left(R_i - \hat{S}_i\right)^2}}{\frac{1}{n_t}\sum_{i=1}^{n_t} R_i} \tag{26}$$

$$\mathrm{PSNR} = 10.\log_{10}\left(\frac{{d_r}^2}{\frac{1}{n_t}\sum_{i=1}^{n_t}\left(R_i - \hat{S}_i\right)^2}\right) = 20.\log_{10}\left(\frac{d_r}{\sqrt{MSE}}\right) \tag{27}$$

where $\mu_{\hat{S}}$ and μ_R, and $\sigma_{\hat{S}}$ and σ_R are the local means and standard deviations of the R and S test data, respectively, and $\sigma_{\hat{S}R}$ is the local cross-covariance. Furthermore, D_1 and D_2 are the nonzero regularization constants and d_r is the dynamic range of pixel brightness. Higher PSNR and SSIM and lower RRMSE values show a more accurate RRN result.

IV. Experimental Results and Discussion

The experimental results of this study are presented in three parts. The first experiment evaluates the efficiency of the FLLF

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TABLE I
PARAMETER VALUES USED IN THE GK-IFLSM SEGMENTATION

Data	constant parameters value				
	μ^{*1}	τ^{*2}	λ^{*3}	ρ^{*4}	ι^{*5}
Dataset 1	0.24	0.84	1.04	4.39	1.56
Dataset 2	0.17	1.16	1.45	6.09	1.12
Dataset 3	0.19	1.07	1.33	5.60	1.22
Dataset 4	0.43	0.46	0.58	2.43	2.81
Dataset 5	0.24	0.83	1.03	4.34	1.57
Dataset 6	0.022	9.08	11.35	47.65	0.14
Dataset 7	0.32	0.63	0.79	0.60	2.05
description	$^{*1}\mu = 1/\varsigma$	$^{*2}\tau = 0.2\varsigma$	$^{*3}\lambda = 0.25\varsigma$		
	$^{*4}\rho = 1.05\varsigma$	$^{*5}\iota = 6.5/\varsigma$			

TABLE II
NUMBER OF THE TEST AND TRAINING DATA FOR THE EXPERIMENTAL DATA SETS

Data	Optimal RCSS	Training samples	Test samples
Dataset 1	34,052	23,866	10,186
Dataset 2	87,154	61,130	26,024
Dataset 3	960,814	672,405	288,409
Dataset 4	398,916	278,899	120,017
Dataset 5	666,851	466,567	200,284
Dataset 6	233,375	163,126	70,249
Dataset 7	1,041,033	728,604	312,429

fusion strategy to produce the difference index. In the second part, the process of selecting the RCSS by the proposed RRN is analyzed. Finally, the third experiment compares the RMSE and RMD of the proposed method against those of several state-of-the-art RRN methods.

Based on previous research [56], the amplification factor and the intensity threshold in this study were set to $S_{\text{factor}} = 1.6$ and $\sigma_r = 0.5$ in the FLLF fusion strategy. For IFLS segmentation, we set $\varepsilon = 1.5$, $\sigma = 2$, and $\nu = 1.70$, as suggested in [31]. Based on the guidelines in [31], [37], [39], and [41], the optimal values of the other IFLSM parameters were calculated by the criterion ς (ratio of area to the length of φ_0) and listed in Table I. The 95% probability contour ($\alpha = 0.05$ level of significance) was considered for the patch-based BVN distribution method. In addition, to minimize the patch effect in this step, we subdivide each of the data sets into 40×40 nonoverlapping blocks. To perform and assess the proposed RRN method, the generated optimal RCSS were randomly split into training data (70%) and testing data (30%) (see Table II).

A. Performance of the FLLF-Based Fusion Strategy

In the first experiment, we appraised the effectiveness of the FLLF fusion strategy to create an acceptable difference index for the RRN. The difference indices generated by the three different operators including CVA, AGD, and FLLF fusion strategy have been shown in Fig. 5. It is clear that the boundaries of the changed regions are easily distinguishable from the background in the AGD index, whereas there is no distinction between the changed and unchanged regions in such an index [see Fig. 5(a)]. This is because the high gradient amplitude highlights the edge or boundary of the changed regions and small local area variation. Regardless of the ability to recognize the changed regions in the CVA index,

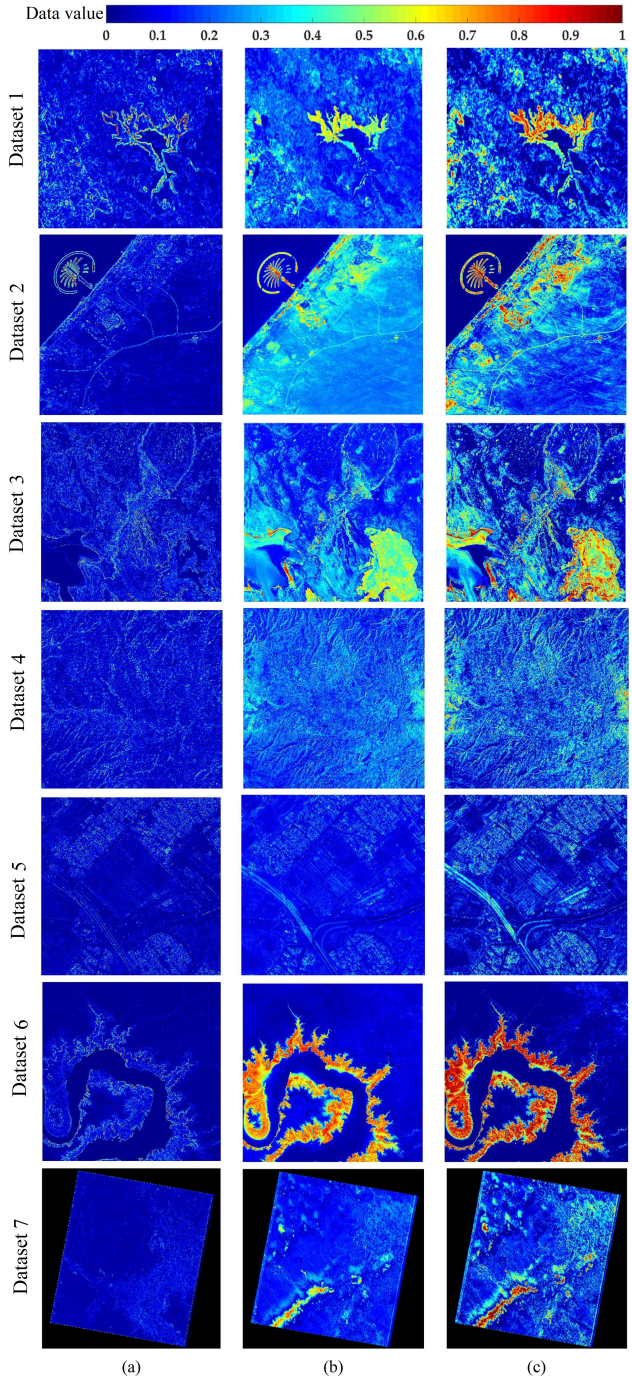

Fig. 5. Generated difference indices of the used data sets by (a) AGD operator, (b) CVA operator, and (c) FLLF fusion strategy.

the weak boundaries of the changed regions and noises will not be significantly distinct and this may hinder the selection of the proper RCSS for the RRN process [see Fig. 5(b)]. Nevertheless, the changed regions, geometry details, and noise are well reflected in the fused difference index generated by the novel FLLF-based strategy for all data sets [see Fig. 5(c)]. This is mainly due to the high sensitivity of the FLLF to the high-frequency values in each of the difference AGD and CVA indices. Moreover, the highlighted samples (e.g., changed regions and isolate points) in the fused difference index will

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TABLE III
AVERAGE RMSE OF SEVEN DATA SETS OBTAINED BY GKC BASED ON THE FIVE DIFFERENCE INDICES

Data	AGD	CVA	OAGD-based	OCVA-based	FLLF-based
Dataset1	4.01	3.47	4.05	3.44	3.37
Dataset2	7.82	6.55	8.41	4.23	4.19
Dataset3	13.43	5.08	14.26	4.84	4.57
Dataset4	6.57	5.27	6.55	5.13	4.98
Dataset5	74.84	66.67	76.11	66.53	65.94
Dataset6	252.39	54.46	280.36	45.94	45.65
Dataset7	13.38	8.05	8.23	7.33	7.25

effortlessly be detected in the process of selecting the RCSS by the desired decision functions. To investigate a role $O_{D_{\text{CVA}}}$ and $O_{D_{\text{AGD}}}$ in the proposed fusion formula, the RRN results of the fused difference index were compared with the ones generated from the CVA and AGD operators, as well as the OAGD-based and OCVA-based as two specific cases of FLLF-based (when $O_{D_{\text{CVA}}} = 0$ and $O_{D_{\text{AGD}}} = 0$), by GKC (see Table III).

From Table III, it can be seen that the proposed fusion strategy sharply outperformed the CVA operator, decreasing the average RMSE of 3%, 36%, 10%, 5%, 1%, 16%, and 10% for data sets 1–7, respectively. This is mainly due to the use of FLLF in the fusion strategy, which enhances the changed regions and their details in the difference index. It is also observed that the AGD operator has a high average RMSE in the normalization process, mainly due to the lack of coverage of the changed areas; thus, it cannot be used alone in the process. By comparing the RRN results of the OAGD-based and OCVA-based by the FLLF-based method, it can be found that the OAGD-based difference index had the worst performance for all data sets except data set 7, mainly because of the limited capability of the AGD operator to cover the changed regions. If no $O_{D_{\text{CVA}}}$ is used at all in the fusion strategy, the average RMSE of the proposed RRN significantly rises to 17%, 50%, 68%, 24%, 13%, 83%, and 12% for data sets 1–7, respectively. Therefore, $O_{D_{\text{CVA}}}$ plays a key and decisive role in the FLLF-based fusion strategy. On the other hand, when $O_{D_{\text{AGD}}} = 0$, the average RMSE of the proposed RRN rises to 6% for data set 3 and 1% for other data sets. The main reason for the significant increase in the RMSE for data set 3 in the absence of $O_{D_{\text{AGD}}}$ is that the geometric details (e.g., the boundary of the changed regions) are not well highlighted in $O_{D_{\text{CVA}}}$, where this deficiency is well covered using $O_{D_{\text{AGD}}}$ in the fusion strategy. Thus, the $O_{D_{\text{AGD}}}$ has only a supporting role in the proposed fusion strategy, and if the changed details in the CVA index are well highlighted, it has not a significant impact on the RRN results. Based on the results above, the FLLF-based fusion strategy can generate a better difference index and it is more suitable to the RRN.

B. Process of Optimization of the Selection of RCSS

In the next phase, the proposed strategy was employed step by step for selecting and optimizing the RCSS. The experimental results of the proposed approach are displayed in Fig. 6(a)–(g).

From Fig. 6, it is clear that the initial contours produced by GKC had been converged to the boundary of the changed regions and the noise-like type one, where the noise-like type two remained in the vicinity of the changed areas in the form of small holes [see Fig. 6(a)]. Although these holes are closed during GK-IFLSM evaluation, the final contours pass from part of the boundary of the changed regions and the noise-like type one [see Fig. 6(b)]. The integration of the initial contour with the final contour takes advantage of both in the detection of the variety of noise-like and changed regions [see Fig. 6(c)]. In fact, such a combination causes small holes in the changed regions to be considered as part of these areas and the noise-like type one, which had been lost during the contour evaluation to be reidentified as unsuitable regions for selecting the RCSS. Therefore, most of the inappropriate regions for selecting the RCSS, including the change regions, isolated points, and types of noise-like, have been eliminated on the change map for all data sets [see Fig. 6(d)]. By comparing this change map with the refined change map (i.e., obtained using the patch-based BVN distribution method), it is observed that a considerable number of unchanged samples have been refined [see Fig. 6(d) and (e)]. Therefore, using the patch-based BVN distribution method had led to the elimination outliers in each patch. In addition, applying this local approach has given us appropriate initial RCSS with well spatial distribution at the whole of the image pair. As can be observed from Fig. 6(f), the histogram shapes do not follow a particular distribution. Therefore, this can justify the use of the median as the core of the proposed automatic thresholding algorithm, which is a distribution-free approach to the evaluation of the dependability of data with complex behaviors. Moreover, the proposed thresholding technique has an effective and defensible performance in RCSS selection due to the optimization of the RCSS based on the proposed purity index in a fully automatic process. This procedure begins slightly (as much as the median coefficient of variation) behind the median of the candidates by initial threshold and ends before their maximum of frequency by the final threshold. Therefore, not only do the selected optimal RCSS have a high degree of purity but also they are evenly distributed throughout the image [see Fig. 6(f) and (g)]. In fact, the spatial distribution and number of such samples within the image appear appropriate such that in each type of unchanged land cover in the data sets (e.g., water body, vegetation, bare soil, building, and road) the optimal RCSS are observed.

To analyze statistically the performance of the proposed method, the normalization coefficients (slope and inception of the OLR) for each spectral band of data sets 1–7 are listed in Table IV.

According to the slopes and intercepts given in Table IV for each spectral band of data sets, the subject images are radiometrically normalized. In addition, the R-squared measures clearly showed that the OLR model provided a very suitable fit for each data set, where the highest and lowest R-squared values are 99.71% and 81.56%, which correspond to the spectral bands 1 and 8 from data sets 2 and 7, respectively. Therefore, it seems that the OLR has succeeded in modeling the relationship between the subject and reference images based on the selected optimal RCSS for all data sets. To estimate the trend of the accuracy of the step-by-step RCSS

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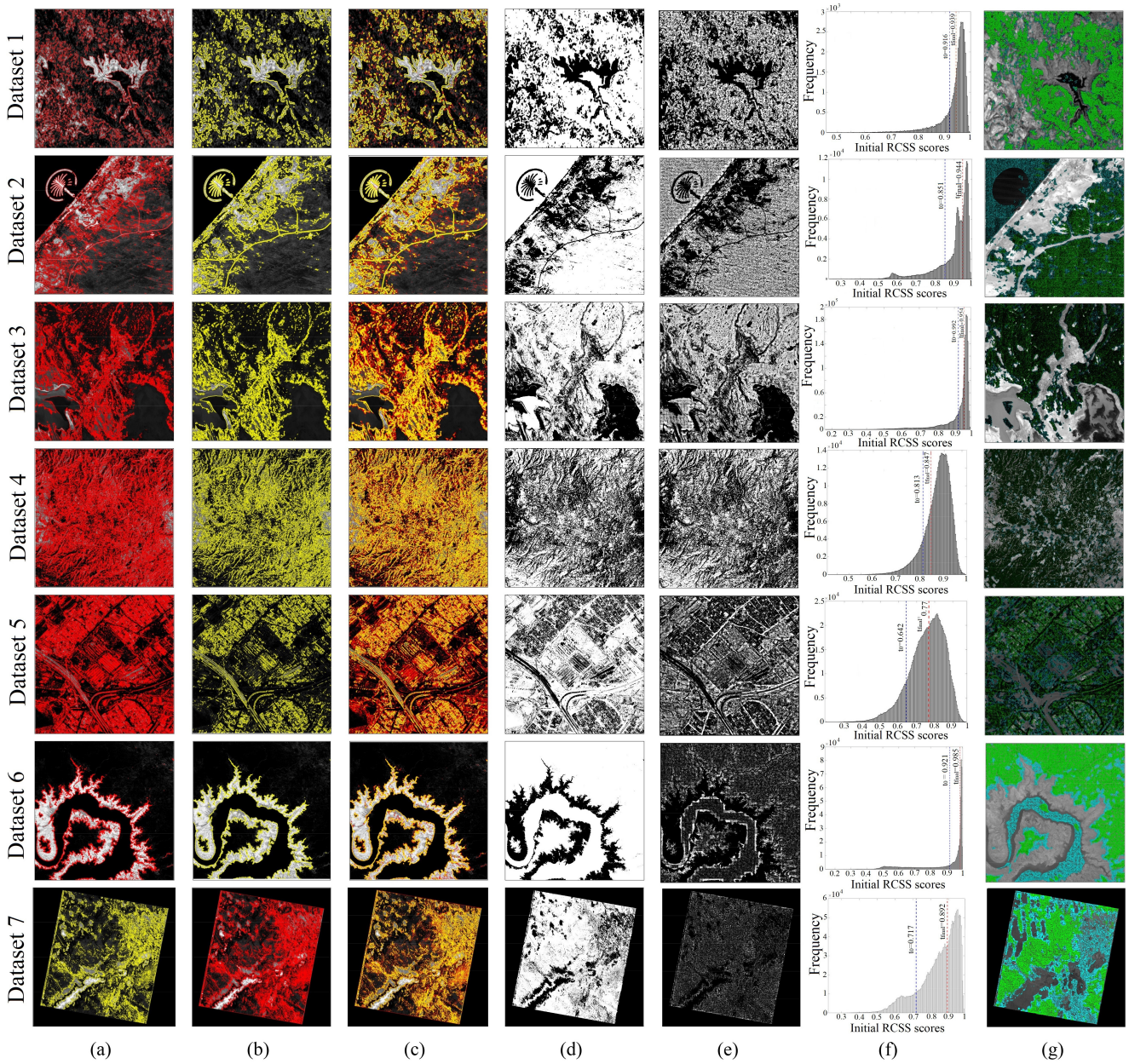

Fig. 6. Experimental results of the proposed step-by-step RCSS selection and optimization for data sets 1–7. (a) Initial contour generated by GKC. (b) Final contour generated by GK-IFLSM through different number of iterations (for data sets 1–7, iterations are 2, 3, 2, 3, 6, 3, and 3, respectively). (c) Overlapping between the initial and final contours. (d) Binary change map generated from combining the GKC and GK-IFLSM results. (e) Initial RCSS maps generated by the patch-based BVN distribution method (white = "initial RCSS"; black = "changed and outlier"). (f) Histograms of the candidate scores, showing the locations of the generated initial and final thresholds. (g) Overlap between the optimal RCSS ("green points") and the initial RCSS ("cyan points").

selection, the unchanged samples generated in each step were considered as RCSS and an effect on the RNN performance was analyzed. The following abbreviations are used to describe the steps for generating the RCSS: applying GKC to D_F (GKC), applying GK-IFLSM to D_F (GK-IFLSM), combining the GKC and GK-IFLSM results (GKC and GK-IFLSM), refining the unchanged samples using the proposed local BVN distribution method (patch-based BVN), and scoring the refined unchanged samples by the PUI and selecting the optimal of them using automatic thresholding (PUI). The RMSE variation trend obtained using the aforementioned steps and their frequency for data sets 1–4 and 7 (with 8-bit radiometric resolution) and data sets 5 and 6 (more than 8-bit radiometric resolution) are displayed in Fig. 7, respectively. Moreover, the regression line corresponding to each step on spectral band 3 of data set 3 is displayed in Fig. 8(a)–(e).

From Fig. 7, a nearly descending RMSE trend in the step-by-step RCSS selection for all of the data sets is observable. It is noteworthy that the RMSE in the second step (GK-IFLSM) is greater than those of the first and third steps (GKC and GKC and GK-IFLSM) for data sets 2, 3, and 7. This is mainly due to the high level of noise-type one (looks like changed regions) in these data sets, which is not considered by the GK-IFLSM and entered into the RRN process as the RCSS. Of course, this problem has been fixed by combining the results of the first and second steps of

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TABLE IV

NORMALIZATION COEFFICIENTS FOR EACH BAND OF THE EXPERIMENTAL DATA SETS

Dataset	Band reference and subject	Normalization coefficients		(R-squared) (%)
		Slope (β_1)	Intercept (β_0)	
Dataset 1	1 and 1	0.914	5.956	95.55
	2 and 2	1.016	8.618	96.85
	3 and 3	1.272	7.1	97.95
	4 and 4	1.217	-1.38	97.13
	5 and 5	1.11	0.816	98.98
	7 and 7	1.121	-3.325	99.62
Dataset 2	1 and 1	0.351	49.118	99.71
	2 and 2	0.873	42.689	99.44
	3 and 3	0.824	46.282	99.13
	4 and 4	0.847	30.093	99.22
	5 and 5	0.648	20.23	99.11
	7 and 7	0.907	18.293	99.18
Dataset 3	1 and 1	0.58	-3.875	97.80
	2 and 2	1.064	-2.236	97.40
	3 and 3	0.928	-1.129	96.23
	4 and 4	0.838	3.392	95.55
	5 and 5	0.625	-0.083	92.57
	7 and 7	0.873	0.722	92.09
Dataset 4	2 and 2	0.869	18.47	99.32
	3 and 3	1.037	16.625	98.99
	4 and 4	0.933	19.654	99.24
Dataset 5	1 and 1	0.5	73.345	98.48
	2 and 2	0.851	40.899	97.46
	3 and 3	0.816	54.868	97.16
	4 and 4	0.807	61.75	96.63
	5 and 5	1.08	35.664	96.42
	6 and 6	0.832	-31.339	97.54
	7 and 7	1.077	11.357	97.97
	8 and 8	0.616	31.697	97.60
Dataset 6	1 and 1	0.888	43.3	99.16
	2 and 2	0.889	20.673	98.54
	3 and 3	0.915	-29.861	97.87
	4 and 4	0.912	-36.87	96.26
	5 and 5	0.931	-60.629	96.05
	6 and 6	0.934	-63.46	95.33
	7 and 7	0.946	-84.137	95.23
	8 and 8	0.97	-114.517	94.88
	8a and 8a	0.97	-123.489	94.78
	9 and 9	1.121	-98.17	94.97
	11 and 11	1.077	-312.667	92.56
	12 and 12	1.078	-242.224	93.85
Dataset 7	1 and 1	0.831	1.462	82.86
	2 and 2	0.788	4.190	85.36
	3 and 3	0.798	1.656	90.32
	4 and 4	0.588	12.741	85.43
	5 and 5	0.587	11.998	86.45
	6 and 6	0.583	13.133	83.86
	7 and 7	0.624	8.451	82.75
	8 and 8	0.642	5.452	81.56
	9 and 9	0.576	9.830	82.38

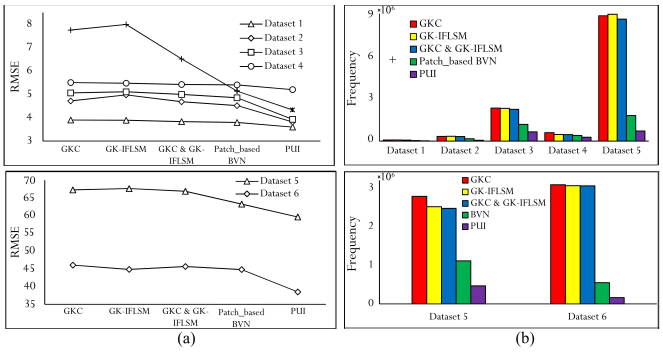

Fig. 7. Experimental RRN results using the step-by-step RCSS selection. (a) Variation trend of RMSE obtained from each step of the RCSS selection by the proposed method. (b) Frequency of each selected RCSS in each step of the proposed RCSS selection and optimization data sets 1–7.

the difference index analysis in the third step (GKC and GK-IFLSM). In total, we observe that the performance of the proposed RCSS selection improves step by step, whereas the

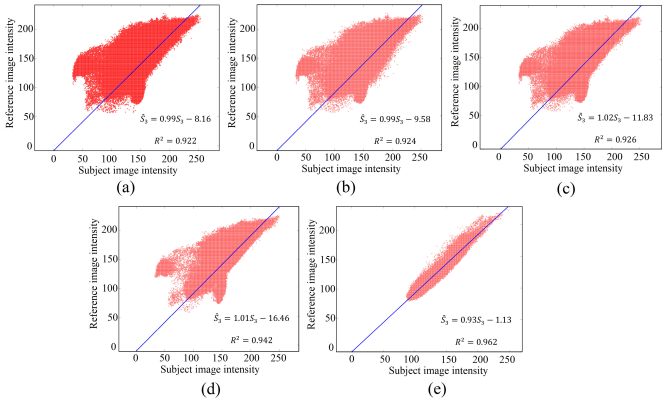

Fig. 8. Regressions on spectral band 3 of data set 3 after the step-by-step RCSS generated by (a) GKC, (b) GK-IFLSM, (c) GKC and GK-IFLSM, (d) patch-based BVN distribution method (BVN), and (e) PUI by automatic thresholding.

frequency gradually decreases for all data sets. Consequently, transitioning step by step through the process of the proposed RCSS selection, satisfactory results are obtained in the final PUI step. This is mainly because the initial RCSS generated by the patch-based BVN method is selected based on their purity values in terms of consistency and being time-invariant. Based on Fig. 8(a)–(e), from left to right, the RCSS that fall beyond the regression line are eliminated from the first to fifth steps. Thus, after each step, the generated RCSS are closer to the fit line than that in the previous step. Moreover, by comparing the R-squared measures of each step, it is clear that the additional PUI step improves the performance of the proposed RCSS selection by reducing the variability of the response variable.

To evaluate the RRN performance, visual interpretation between the subject, reference, and normalized images is done [see Fig. 4(a)–(c)]. According to Fig. 4, the brightness and contrast of the normalized images are closer to those in the reference images, whereas the original subject images have noticeable contrast and low brightness. Therefore, the original subject images and the corresponding reference images are visually different and, in particular, the spectral difference of the diverse unchanged land covers (e.g., vegetation, bare soil, road, water body, and so on) is relatively high due to the time interval between the acquisitions of the image pair. In general, the spectral differences between the reference and normalized subject images are significantly reduced, especially in terms of unchanged vegetation though there are qualitative changes in the vegetation (changes in the chemical composition of plants such as plant color and genus changes) due to the acquisition of the subject and reference images in different seasons and years. Moreover, the areas of the subject image, which are covered by the cloud in the reference image, have been well enhanced in the normalized image generated, which can be considered as an important achievement in the field of radiometric correction by the proposed method (see Fig. 4, last row).

The difference histograms of the reference and subject images $[d_{(r-s)}]$ were compared with those of the reference and normalized subject images $[d_{(r-ns)}]$ in the NIR spectral band

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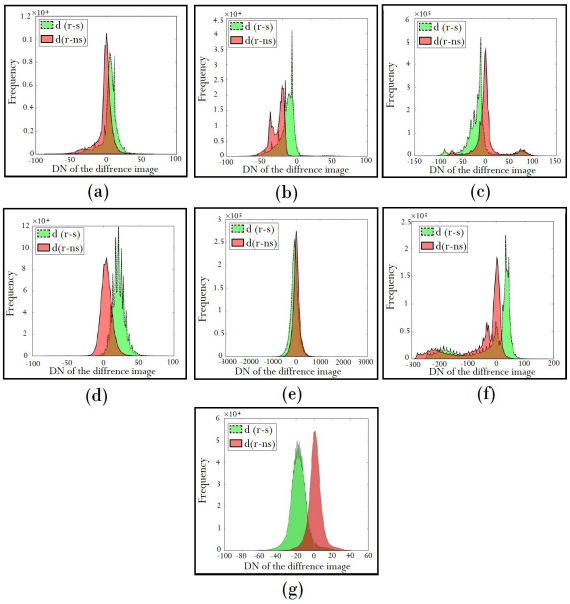

Fig. 9. Comparison between histogram $d_{(r-s)}$ and $d_{(r-ns)}$ in the NIR band for (a) data set 1, (b) data set 2, (c) data set 3, (d) data set 4, (e) data set 5, (f) data set 6, and (g) data set 7.

for the data sets [see Fig. 9(a)–(g)]. As shown in Fig. 9, the histograms of $d_{(r-ns)}$ have Gaussian shape and are narrower than that of $d_{(r-s)}$ for data sets 1, 3, 4, 5, and 7. This is because the number of pixels located at zero and near-zero position (unchanged part) has increased in $d_{(r-ns)}$. In data sets 2 and 6, which include large water body changes, the conditions are slightly more complicated. In these data sets, the histograms of $d_{(r-ns)}$ have more than one peak (due to the diversity of the land-cover changes), but each almost follows a Gaussian distribution, where the histograms of $d_{(r-s)}$ are less defined [see Fig. 9(b) and (f)]. The main peaks of the difference histograms $d_{(r-ns)}$ are centered around the zero value and have far less standard deviation. This is an advantage of using the RCSS in the RNN, which are selected and optimized by the proposed step-by-step approach.

The comparison between the means and standard deviations of the reference, subject, and normalized images is presented in Fig. 10. As shown in Fig. 10, the MVs for all the spectral bands of the normalized subject images were significantly closer to those of the reference images. Moreover, the differences in the standard deviation between the reference and normalized images are less than those between the reference and subject images for all spectral bands except bands 6–8 in data set 5 and band 4 in data set 6. This is because there is a noteworthy difference between the DNs of some of the test pixels of the aforementioned spectral bands in data sets 5 and 6. In general, the minor differences between the mean and standard deviations of the normalized and reference images indicate that the proposed method adequately reduces the radiometric distortions from the original subject images.

Fig. 11, compares the SSIM, PSNR, and RRMSE values of the reference and subject images with those of the reference and normalized subject images in the test samples. A cursory look at Fig. 11(a)–(c) reveals that the PSNR (dB) and SSIM

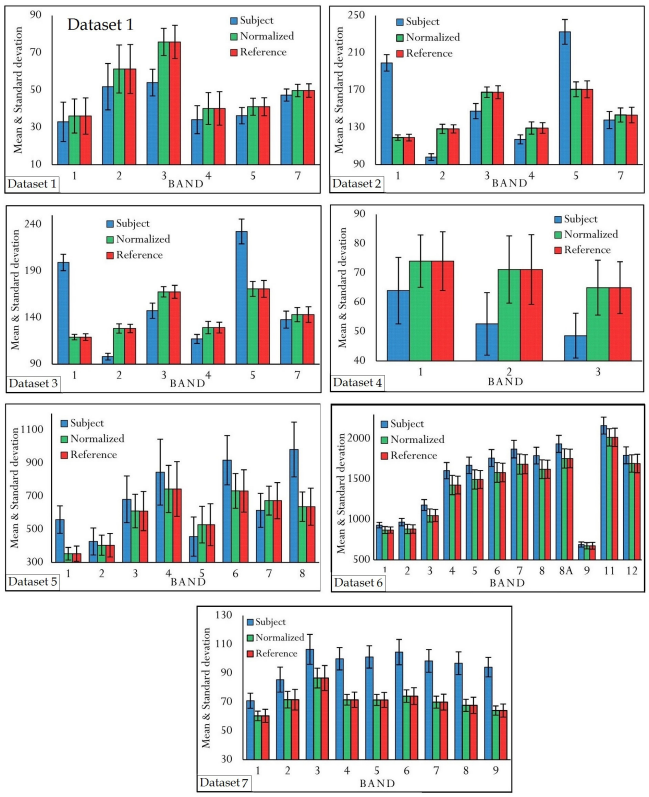

Fig. 10. Validation of mean and standard deviation in all spectral bands on the check samples for data sets 1–7.

values of all spectral bands increased, while the RRMSE values were considerably reduced after the normalization of all data sets. Given this, the performance of the proposed method was different for the various spectral bands of each data set. The proposed RRN method has the greatest efficacy on spectral bands 3, 1, 1, 2, 1, 4, and 9 for data sets 1–7, respectively, where the differences in the SSIM, PSNR (dB), and RRMSE values before and after normalization were virtually at the maximum. Furthermore, the proposed RRN method has the least impact on spectral bands 7, 7, 2, 1, 2, 9, and 1 for data set 1–7, respectively, where the differences in the SSIM, PSNR (dB), and RRMSE values between before and after normalization were nearly minimum. The best performance of the proposed method is observed in data sets 3 and 7, where the differences in the three objective quantity values before and after normalization were significantly high in all the spectral bands. Therefore, the proposed method has a high potential in the RRN of the data set with a large number of land-cover changes.

V. Comparing the Performance of RRN Methods

To demonstrate the capability of the proposed RRN method, we compare it with the following prevalent state-of-the-art approaches: 1) HC: haze correction [19]; 2) HM: histogram matching [20]; 3) MM: minimum–maximum [14]; 4) MS: mean–standard deviation [14]; 5) SR: simple regression [18]; 6) DB: dark set–bright set based on the Kauth–Thomas greenness-brightness transformation [14]; 7) PIF: pseudoinvariant feature approach [23]; 8) MPIF: normalizing

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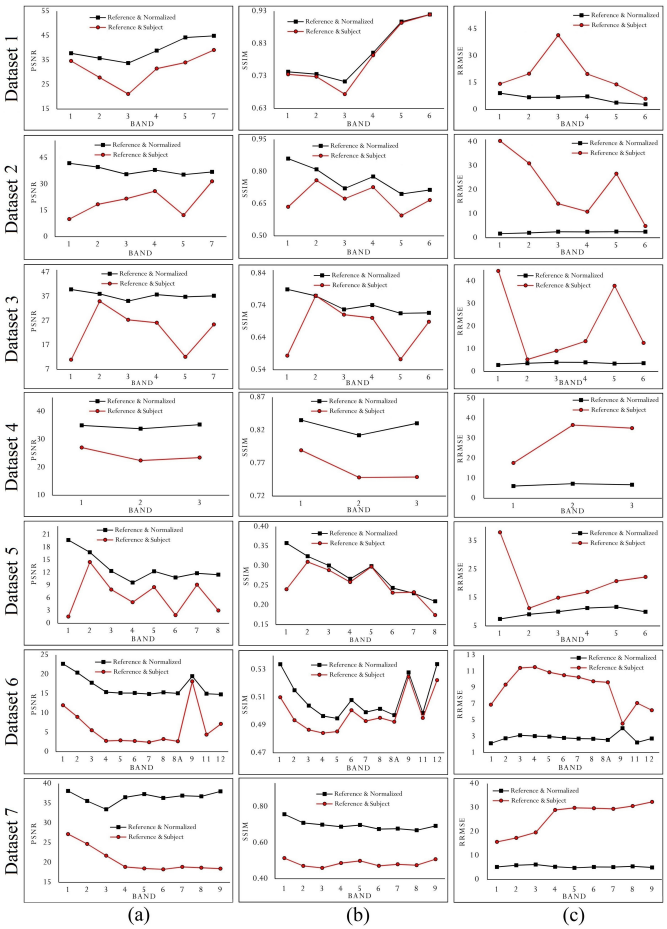

Fig. 11. Comparison of (a) PSNR (dB), (b) SSIM, and (c) RRMSE (%) values of the reference and subject images with those of the reference and normalized subject images in all spectral bands of data sets.

multiple images with PIFs [17]; 9) UC-RF: normalizing image based on RF regression and unchanged pixels [30]; 10) PCA-based: principle component analysis-based correction [25]; 11) ARIN: semiautomatic normalization image based on the vegetation PIFs (VPIFs) [21]; 12) UC-LR: normalizing image employed by a linear regression and unchanged pixels, i.e., dark, gray, and bright categories [26]; 13) UC-ANN: normalizing image based on an ANN and unchanged pixels [16]; 14) MAD: multivariate alteration-detection transformation-based normalization [27]; 15) HOG-based: normalizing image based on the HOG features as the RCSS [22]; 16) HOGC-based: normalizing image based on the feature category, i.e., HOG features in each of the class that is obtained by a supervised object-oriented classification (SOC) method [22]; and 17) IR-MAD: iteratively reweighted MAD (IR-MAD) transformation-based normalization [29].

The DB method has primarily been applied to the Landsat images (data sets 1–3). The thresholds of the DB and PIF methods were selected using a trial-and-error approach. According to [16], for the UC-ANN, one hidden layer was considered in the ANN architecture with the appropriate number of neurons (i.e., selected based on the trial-and-error approach from 1 to 10 with 300 repetition rates for each spectral band in data sets 1–7). In the ARIN method, the unchanged vegetation samples were manually extracted as VPIFs for normalizing the subject image. The other approaches were employed using the same set of parameters as offered in the corresponding articles. The experimental results obtained from the aforementioned methods for data sets 1–7 are summarized in Table V.

A cursory glance at Table V confirms that superior normalization results were obtained from the SRRN methods, including the proposed method, IR-MAD, MAD, UC-LR, NC-ANN, PCA-based, MIPIF, PIF, DB, NC-RF, HOG-based, and HOG-based methods. The proposed method provides superior results overall, followed closely by IR-MAD, which is in turn followed by MAD. Applying the proposed method improved the raw average RMSE values by 64.35%, 90.66%, 88.34%, 70.38%, 62.02%, 72.72%, and 84.77% for data sets 1–7, respectively. Furthermore, the mean RMD values were enhanced by 66.55%, 93.15%, 90.83%, 75.34%, 69.84%, 76.77%, and 88.36% for data sets 1–7, respectively. The performance difference between the proposed and IR-MAD methods is more obvious in data sets 4 and 6. This is because the IR-MAD method selects the RCSS based only on the probability of the unchanged samples in the transformed image space instead of the original image space. In addition, the performance of the MAD method improved with an iterative procedure under the format of the IR-MAD method. The HOG-based method had a relatively feeble performance among the SRRN methods. This is because although it selects unchanged pixels as the RCSS, the same normalization coefficients are used for different land-cover classes in the reference-subject image pairs, which is not sensible. Undeniably, this problem has been partly resolved by improving the concept of the HOG-based method by adding a supervised object-oriented classification when applied to the data sets used herein. However, the performance of HOGC-based is dependent on the used image classification algorithm and its associated accuracy. As expected, the ARIN method performed acceptably in the data sets containing a large amount of vegetation, such as data sets 1 and 4, although destroyed the normalization results in regions with a variety of land covers or with low vegetation, such as data sets 2, 3, 5, 6, and 7. This is because only vegetation RCSS were used for normalization. The MIPIF, UC-RF, and UC-ANN methods have nearly the same overall performance, although these methods behave differently in each data set. Even though the UC-ANN had a relatively good performance when applied to data sets 1–4 and 7, it severely distorted the normalization results in data sets 5 and 6. This is because ANN modeling is highly sensitive to the outliers and the existence of such samples has led to the falsification of the RRN results for data sets 5 and 6. The UC-RF method had a relatively balanced performance, although the normalization results are slightly biased in the data sets with the considerable land-cover change, such as data sets 4–6. This can be related to the assumptions of this method (the amount of changed areas and the correlation coefficient threshold setting that may cause some time-invariant samples to be overlooked in the RRN with this method). By examining the performance of the MIPIF method, we found that it is more dependent on the presence of unchanged water body areas. This method works fairly well in

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TABLE V
AVERAGE RMSE AND RMD OVER THE SPECTRAL BANDS RESULTING FROM DIFFERENT RRN METHODS APPLIED TO DATA SETS 1–7

No.	RRN Methods	Dataset 1		Dataset 2		Dataset 3		Dataset 4		Dataset 5		Dataset 6		Dataset 7	
		RMSE	RMD	RMSE	RMD	RMSE	RMD	RMSE	RMD	RMSE	RMD	RMSE	RMD	RMSE	RMD
0	Raw	8.696	14.555	35.527	25.433	29.423	30.228	15.875	21.724	157.028	25.603	141.113	9.489	25.170	34.861
1	HC [19]	9.443	15.851	25.998	17.551	32.687	33.66	10.699	14.206	155.509	25.685	101.488	6.21	25.374	35.250
2	HM [20]	5.879	8.013	7.712	3.926	6.206	4.503	7.103	8.107	98.31	13.424	86.482	4.936	5.676	5.586
3	MM [14]	7.263	12.073	14.904	11.749	16.007	20.909	11.406	16.23	119.545	18.279	173.117	12.872	25.699	35.718
4	MS [14]	4.418	7.587	13.999	8.109	14.323	12.055	6.598	7.533	93.883	12.799	70.592	4.132	16.909	22.466
5	SR [18]	4.145	6.853	8.712	5.553	14.134	13.664	5.891	6.678	75.689	9.625	87.209	4.984	5.312	5.975
6	DB [14]	5.114	8.608	5.486	2.655	8.19	7.61	—	—	—	—	—	—	—	—
7	PIF [23]	4.171	6.972	5.93	2.845	7.276	6.617	5.528	6.345	77.92	9.98	46.992	2.858	9.325	11.607
8	MIPIF [17]	3.682	5.681	4.242	2.232	4.219	3.478	6.583	13.578	73.007	9.507	42.646	2.517	7.815	9.323
9	UC-RF [30]	3.319	5.187	4.966	2.836	3.804	2.954	6.686	7.596	65.793	8.463	45.049	2.508	4.319	4.557
10	PCA-based [25]	3.436	5.235	4.395	2.465	4.648	4.164	6.973	7.895	66.048	8.57	42.38	2.524	4.796	5.288
11	ARIN [21]	3.409	5.712	36.764	25.769	15.001	14.996	6.981	9.396	103.217	16.106	191.445	11.908	15.471	20.310
12	UC-LR [26]	3.821	5.648	4.576	2.573	5.407	4.9	6.845	7.801	74.769	9.55	45.977	2.746	5.110	5.905
13	UC-ANN [16]	3.444	5.324	3.549	1.79	4.779	3.612	4.887	5.646	93.694	13.288	44.441	2.42	4.491	4.824
14	MAD [27]	3.217	5.101	4.22	2.407	3.735	2.935	7.137	8.228	61.47	7.955	40.876	2.411	4.255	4.551
15	HOG-based [22]	3.883	6.139	7.344	3.901	5.064	4.383	7.357	8.67	68.064	8.86	66.406	4.073	5.707	6.544
16	HOGC-based [22]	3.757	5.964	4.263	2.429	3.995	3.387	6.382	7.271	66.285	8.519	41.266	2.381	4.643	5.191
17	IR-MAD [29]	3.187	5.005	3.847	2.177	3.727	2.92	6.972	7.983	60.617	7.859	40.298	2.37	4.229	4.523
18	Proposed method	3.100	4.869	3.319	1.742	3.432	2.771	4.702	5.357	59.634	7.721	38.498	2.204	3.833	4.059

data sets 1, 2, and 6, which have unchanged water body zones, whereas it does not include an unchanged water body, which is because this method uses the NDWI to select the RCSS. The PCA-based and UC-LR methods had oscillating behavior in the RRN of the data sets. These methods had a relatively moderate performance in data sets 1, 3, and 6, which have low radiometric differences. However, their performance is deeply influenced by the data sets with high radiometric deformations, like data sets 4 and 5. The DB and PIF methods have mediocre performance, where their thresholds were adapted based on the accumulated knowledge from experience. The DRRN methods had unsatisfactory results compared with the SRRN methods for all experimented data sets. All these methods, including HM, HC, MM, MS, and SR methods, use global image statistics without considering the changed/unchanged regions to adjust the radiometric property of the subject images, and therefore, their performance suffers. That is why it was expected that using the RCSS in the normalization process would outperform the DRRN methods.

VI. CONCLUSION

In this research, we introduced an improved SRRN method by exploiting a new difference index and the step-by-step RCSS selection and optimization method to address the limitations that exist in the RRN area. This research was conducted on seven different bitemporal satellite images, taken under as many different conditions from a variety of land-cover changes. The experimental results determined that the proposed RCSS selection method was reliable and the fused difference index dealt with the efficiency of each of these CVA and AGD difference indices so that the unchanged region is well distinguished from the changed regions. In addition, a combination of both GKC and GK-IFLSM change maps leads to a significant improvement in the RRN results compared with both GK and GK-IFLSM, where noise-like structures were removed in the final change map. Applying the patch-based BVN distribution method is effective for identifying outliers from the unchanged samples and reducing the search space for determining the initial RCSS. Comparing the SSIM, PSNR, and RRMSE values of the reference and subject images with those of the reference and normalized subject images in the check samples demonstrates that the differences between the reference and subject images decreased considerably after the normalization in terms of quantitative and visual characteristics. Moreover, the difference histogram of the reference and normalized subject images had a normal Gaussian distribution by a lesser standard deviation around the zero-position than those of the reference and original subject images. One of the proficiencies of the proposed method is that the purity degrees of the initial RCSS are determined, not only based on the no-change probability but also their perceptual qualities (i.e., their luminance, contrast, and structure). The proposed iterative thresholding selected optimal RCSS in a fully automatic procedure that is free of any distributions about modeling the pdfs of the initial RCSS. Our extensive evaluations on different satellite data sets verify that the normalized image of the proposed RRN method is superior to those of some used existing RRN methods in terms of RMSE and RMD. Importantly, the proposed method has a high generality when dealing with different satellite data sets with diverse land covers. Overall, the effective RRN strategy proposed herein will be extremely useful for a variety of satellite imagery applications in numerous disciplines.

REFERENCES

[1] P. M. Teillet, "Image correction for radiometric effects in remote sensing," *Int. J. Remote Sens.*, vol. 7, no. 12, pp. 1637–1651, Dec. 1986.
[2] X. Yang and C. P. Lo, "Relative radiometric normalization performance for change detection from multi-date satellite images," *Photogramm. Eng. Remote Sens.*, vol. 66, no. 8, pp. 967–980, Aug. 2000.
[3] F. Hui, B. Xu, H. Huang, Q. Yu, and P. Gong, "Modelling spatial-temporal change of poyang lake using multitemporal Landsat imagery," *Int. J. Remote Sens.*, vol. 29, no. 20, pp. 5767–5784, 2008.
[4] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2007.
[5] A. Ghosh, N. S. Mishra, and S. Ghosh, "Fuzzy clustering algorithms for unsupervised change detection in remote sensing images," *Inf. Sci.*, vol. 181, no. 4, pp. 699–715, Feb. 2011.

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[6] S. Vicenteserrano, F. Perezcabello, and T. Lasanta, "Assessment of radiometric correction techniques in analyzing vegetation variability and change using time series of landsat images," *Remote Sens. Environ.*, vol. 112, no. 10, pp. 3916–3934, Oct. 2008.

[7] Y.-H. Tu, S. Phinn, K. Johansen, and A. Robson, "Assessing radiometric correction approaches for multi-spectral UAS imagery for horticultural applications," *Remote Sens.*, vol. 10, no. 11, p. 1684, Oct. 2018.

[8] Y. Zhang, L. Yu, M. Sun, and X. Zhu, "A mixed radiometric normalization method for mosaicking of high-resolution satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2972–2984, May 2017.

[9] W. Chen, K. Moriya, T. Sakai, L. Koyama, and C. X. Cao, "Mapping a burned forest area from Landsat TM data by multiple methods," *Geomatics, Natural Hazards Risk*, vol. 7, no. 1, pp. 384–402, Jan. 2016.

[10] M. Schultz, J. Verbesselt, V. Avitabile, C. Souza, and M. Herold, "Error sources in deforestation detection using BFAST monitor on Landsat time series across three tropical sites," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3667–3679, Aug. 2016.

[11] Q. Xu, Z. Hou, and T. Tokola, "Relative radiometric correction of multitemporal ALOS AVNIR-2 data for the estimation of forest attributes," *ISPRS J. Photogramm. Remote Sens.*, vol. 68, pp. 69–78, Mar. 2012.

[12] N. Yan, B. Wu, V. K. Boken, S. Chang, and L. Yang, "A drought monitoring operational system for China using satellite data: Design and evaluation," *Geomatics, Natural Hazards Risk*, vol. 7, no. 1, pp. 264–277, Jan. 2016.

[13] R. Gens, "Remote sensing of coastlines: Detection, extraction and monitoring," *Int. J. Remote Sens.*, vol. 31, no. 7, pp. 1819–1836, Apr. 2010.

[14] F. G. Hall, D. E. Strebel, J. E. Nickeson, and S. J. Goetz, "Radiometric rectification: Toward a common radiometric response among multidate, multisensor images," *Remote Sens. Environ.*, vol. 35, no. 1, pp. 11–27, Jan. 1991.

[15] N. Bao, A. M. Lechner, A. Fletcher, D. Mulligan, A. Mellor, and Z. Bai, "Comparison of relative radiometric normalization methods using pseudo-invariant features for change detection studies in rural and urban landscapes," *J. Appl. Remote Sens.*, vol. 6, no. 1, 2012, Art. no. 063578.

[16] V. Sadeghi, H. Ebadi, and F. F. Ahmadi, "A new model for automatic normalization of multitemporal satellite images using artificial neural network and mathematical methods," *Appl. Math. Model.*, vol. 37, no. 9, pp. 6437–6445, May 2013.

[17] H. Zhou, S. Liu, J. He, Q. Wen, L. Song, and Y. Ma, "A new model for the automatic relative radiometric normalization of multiple images with pseudo-invariant features," *Int. J. Remote Sens.*, vol. 37, no. 19, pp. 4554–4573, Oct. 2016.

[18] J. R. Jensen, "Urban/suburban land use analysis," in *Manual of Remote Sensing*, 2nd ed. Falls Church, VA, USA: Wiely, 1983, pp. 1571–1666.

[19] P. S. Chavez, "An improved dark-object subtraction technique for atmospheric scattering correction of multispectral data," *Remote Sens. Environ.*, vol. 24, no. 3, pp. 459–479, Apr. 1988.

[20] J. A. Richards and J. Richards, *Remote Sensing Digital Image Analysis*, vol. 3. Berlin, Germany: Springer, 1999.

[21] L. Garcia-Torres, J. J. Caballero-Novella, D. Gómez-Candón, and A. I. De-Castro, "Semi-automatic normalization of multitemporal remote images based on vegetative pseudo-invariant features," *PLoS ONE*, vol. 9, no. 3, Mar. 2014, Art. no. e91275.

[22] H. He, X. Liu, and Y. Shen, "Relative radiometric correction of high-resolution remote sensing images based on feature category," *Cluster. Comput.*, vol. 22, no. 3, pp. 1–9, 2018.

[23] C. Salvaggio, "Radiometric scene normalization utilizing statistically invariant features," in *Proc. Workshop Atmos. Correction Landsat Imag.*, Torrance, CA, USA, 1993, pp. 155–159.

[24] R. J. Kauth and G. Thomas, "The tasselled cap—A graphic description of the spectral-temporal development of agricultural crops as seen by landsat," in *Proc. LARS Symp.*, 1976, p. 159.

[25] Y. Du, P. M. Teillet, and J. Cihlar, "Radiometric normalization of multitemporal high-resolution satellite images with quality control for land cover change detection," *Remote Sens. Environ.*, vol. 82, no. 1, pp. 123–134, Sep. 2002.

[26] S. M. Ya'allah and M. R. Saradjian, "Automatic normalization of satellite images using unchanged pixels within urban areas," *Inf. Fusion*, vol. 6, no. 3, pp. 235–241, Sep. 2005.

[27] M. J. Canty, A. A. Nielsen, and M. Schmidt, "Automatic radiometric normalization of multitemporal satellite imagery," *Remote Sens. Environ.*, vol. 91, nos. 3–4, pp. 441–451, Jun. 2004.

[28] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, Apr. 1998.

[29] M. J. Canty and A. A. Nielsen, "Automatic radiometric normalization of multitemporal satellite imagery with the iteratively re-weighted MAD transformation," *Remote Sens. Environ.*, vol. 112, no. 3, pp. 1025–1036, Mar. 2008.

[30] D. Seo, Y. Kim, Y. Eo, W. Park, and H. Park, "Generation of radiometric, phenological normalized image based on random forest regression for change detection," *Remote Sens.*, vol. 9, no. 11, p. 1163, Nov. 2017.

[31] A. Moghimi, S. Khazai, and A. Mohammadzadeh, "An improved fast level set method initialized with a combination of k-means clustering and Otsu thresholding for unsupervised change detection from SAR images," *Arabian J. Geosci.*, vol. 10, no. 13, p. 293, Jul. 2017.

[32] S. Paris, S. W. Hasinoff, and J. Kautz, "Local Laplacian filters: Edge-aware image processing with a Laplacian pyramid," *ACM Trans. Graph.*, vol. 30, no. 4, pp. 1–68, 2011.

[33] P. Burt and E. Adelson, "The Laplacian pyramid as a compact image code," *IEEE Trans. Commun.*, vol. TCOMM-31, no. 4, pp. 532–540, Apr. 1983.

[34] M. Aubry, S. Paris, S. W. Hasinoff, J. Kautz, and F. Durand, "Fast local Laplacian filters: Theory and applications," *ACM Trans. Graph.*, vol. 33, no. 5, pp. 167:1–167:14, Sep. 2014.

[35] S. Yu and W. Yiquan, "An improved local Laplacian filter based on the relative total variation," *Digit. Signal Process.*, vol. 78, pp. 56–71, Jul. 2018.

[36] Z. Qiang, L. He, Y. Chen, X. Chen, and D. Xu, "Adaptive fast local Laplacian filters and its edge-aware application," *Multimedia Tools Appl.*, vol. 78, pp. 619–639, Nov. 2017.

[37] C. Li, C. Xu, C. Gui, and M. D. Fox, "Level set evolution without re-initialization: A new variational formulation," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun. 2005, pp. 430–436.

[38] S. Osher and J. A. Sethian, "Fronts propagating with curvature-dependent speed: Algorithms based on Hamilton-Jacobi formulations," *J. Comput. Phys.*, vol. 79, no. 1, pp. 12–49, Nov. 1988.

[39] B. N. Li, C. K. Chui, S. Chang, and S. H. Ong, "Integrating spatial fuzzy clustering with level set methods for automated medical image segmentation," *Comput. Biol. Med.*, vol. 41, no. 1, pp. 1–10, Jan. 2011.

[40] C. Xu and J. L. Prince, "Snakes, shapes, and gradient vector flow," *IEEE Trans. Image Process.*, vol. 7, no. 3, pp. 359–369, Mar. 1998.

[41] A. Moghimi, A. Mohammadzadeh, and S. Khazai, "Integrating thresholding with level set method for unsupervised change detection in multitemporal SAR images," *Can. J. Remote Sens.*, vol. 43, no. 5, pp. 412–431, Sep. 2017.

[42] A. Sedaghat, M. Mokhtarzade, and H. Ebadi, "Uniform robust scale-invariant feature matching for optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4516–4527, Nov. 2011.

[43] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. LARS Symp.*, 1980, p. 385.

[44] D. Gustafson and W. Kessel, "Fuzzy clustering with a fuzzy covariance matrix," in *Proc. IEEE Conf. Decis. Control Including 17th Symp. Adapt. Processes*, Jan. 1978, pp. 761–766.

[45] S. P. Teng, Y. K. Chen, K. S. Cheng, and H. C. Lo, "Hypothesis-test-based landcover change detection using multi-temporal satellite images—A comparative study," *Adv. Space Res.*, vol. 41, no. 11, pp. 1744–1754, Jan. 2008.

[46] P. J. Rousseeuw and K. V. Driessen, "A fast algorithm for the minimum covariance determinant estimator," *Technometrics*, vol. 41, no. 3, pp. 212–223, Aug. 1999.

[47] A. M. Leroy and P. J. Rousseeuw, *Robust Regression and Outlier Detection*. New York, NY, USA: Wiley, 1987.

[48] C. Leys, O. Klein, Y. Dominicy, and C. Ley, "Detecting multivariate outliers: Use a robust variant of the mahalanobis distance," *J. Exp. Social Psychol.*, vol. 74, pp. 150–156, Jan. 2018.

[49] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[50] M. Kendall and A. Stuart, "The advanced theory of statistics," in *Distribution Theory*, vol. 1, 4th ed. London, U.K.: Griffin, 1977.

[51] T. Keleş and M. Altun, "Comparison of classical linear regression and orthogonal regression according to the sum of squares perpendicular distances," *Eğitimde ve Psikolojide Ölçme ve Değerlendirme Dergisi*, vol. 7, no. 2, pp. 296–308, Dec. 2016.

[52] *Awesome Remote Sensing Relative Radiometric Normalization-Datasets, Retrieved on May 2020 from the World Wide Web. Github Platform*. Accessed: May 14, 2020. [Online]. Available: https://github.com/ArminMoghimi/Awesome-Remote-Sensing-Relative-Radiometric-Normalization-Datasets

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[53] R. Welch, "Merging multiresolution SPOT HRV and landsat TM data," *Photogramm. Eng. Remote. Sens.*, vol. 53, no. 3, pp. 301–303, 1987.
[54] N. Brodu, "Super-resolving multiresolution images with band-independent geometry of multispectral pixels," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4610–4617, Aug. 2017.
[55] *Retrieved on January 2018 From the World Wide Web. ESA Scientific Toolbox Exploitation Platform*. Accessed: Jun. 1, 2020. [Online]. Available: http://step.esa.int/main/third-party-plugins-2/sen2res/
[56] N.-D. Hoang and Q.-L. Nguyen, "Fast local Laplacian-based steerable and Sobel filters integrated with adaptive boosting classification tree for automatic recognition of asphalt pavement cracks," *Adv. Civil Eng.*, vol. 2018, Sep. 2018, Art. no. 5989246.

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